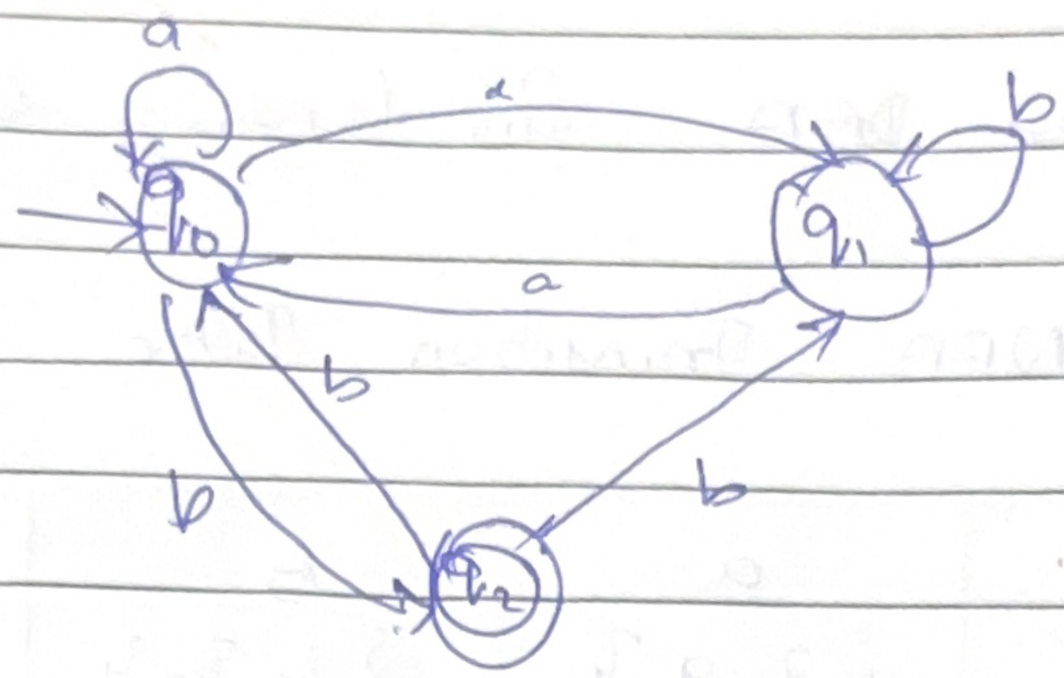
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① Consider the NFA given below which accept some language L over the alphabet $\Sigma = (a, b)$. Construct a DFA the represents the same language L



States : Q_0, Q_1, Q_2

Start state : q_0

Final state : $f = \{q_2\}$

State	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	$\{\}$	$\{q_1, q_0\}$

① Start state $A = \{q_0\}$

② Transition from A ($\{q_0\}$)

③ Transitions from B ($\{q_0, q_1\}$)

④ Transition from C ($\{q_2\}$)

$\delta_{DFA}(C, a) = \delta(q_2, a) = \phi = E$

⑤ Transition from D ($\{q_1, q_2\}$)

⑥ Transition from E (ϕ)

DFA State	a	b
A (q_0)	B	C
B (q_0, q_1)	B	D
* C (q_2)	E	B
* D (q_1, q_2)	A	B
E (ϕ)	E	E

2. NFA and DFA for language L

1. NFA Transition Table

State	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_2	ϕ	$\{q_3\}$
q_3	$\{q_3\}$	ϕ

2. Equivalent DFA construction

$S_0 \Rightarrow \{q_0\}$ Start

$S_1 = \{q_0, q_1\}$

$S_2 = \{q_0, q_2\}$

$S_3 = \{q_0, q_1, q_3\}$

State (subset)	a	b
→ S_0 $\{q_0\}$	S_1	S_2
S_1 $\{q_0, q_1\}$	S_1	S_3
S_2 $\{q_0, q_2\}$	S_4	S_2
S_3 $\{q_0, q_1, q_3\}$	S_4	S_5
S_4 $\{q_0, q_1, q_3\}$	S_1	S_3